Introduction to Artificial intelligence

Tutorial Sheet 6 (Logic and Reasoning - Solution)

Exercise 1:

Consider a vocabulary with only four propositions, *A*, *B*, *C*, and *D*. How many interpretations (i.e. possible worlds) and how many models are there for the following sentences?

a. $B \vee C$.
b. $\neg A \vee \neg B \vee \neg C \vee \neg D$.
c. $(A \Rightarrow B) \wedge A \wedge \neg B \wedge C \wedge D$.
d. $(A \wedge B) \vee (C \wedge D)$.
e. $B \Rightarrow (A \wedge B)$.

Exercise 2:

Decide whether each of the following sentences is valid, unsatisfiable, or neither. Verify your decisions using truth tables or the equivalence rules of Figure 7.11 (page 223).

a. $Smoke \Rightarrow Smoke$
b. $Smoke \Rightarrow Fire$
c. $(Smoke \Rightarrow Fire) \Rightarrow (\neg Smoke \Rightarrow \neg Fire)$
d. $Smoke \vee Fire \vee \neg Fire$
e. $((Smoke \wedge Heat) \Rightarrow Fire) \Leftrightarrow ((Smoke \Rightarrow Fire) \vee (Heat \Rightarrow Fire))$
f. $(Smoke \Rightarrow Fire) \Rightarrow ((Smoke \wedge Heat) \Rightarrow Fire)$
g. $Big \vee Dumb \vee (Big \Rightarrow Dumb)$

$$
\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{De Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{De Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}
$$

Figure 7.11 Standard logical equivalences. The symbols α, β, and γ stand for arbitrary sentences of propositional logic.

Exercise 3:

Which of the following are correct?

a. $False \models True$.
b. $True \models False$.
c. $(A \wedge B) \models (A \Leftrightarrow B)$.
d. $A \Leftrightarrow B \models A \vee B$.
e. $A \Leftrightarrow B \models \neg A \vee B$.
f. $(A \wedge B) \Rightarrow C \models (A \Rightarrow C) \vee (B \Rightarrow C)$.
g. $(C \vee (\neg A \wedge \neg B)) \equiv ((A \Rightarrow C) \wedge (B \Rightarrow C))$.
h. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B)$.
i. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B) \wedge (\neg D \vee E)$.
j. $(A \vee B) \wedge \neg(A \Rightarrow B)$ is satisfiable.
k. $(A \Leftrightarrow B) \wedge (\neg A \vee B)$ is satisfiable.
l. $(A \Leftrightarrow B) \Leftrightarrow C$ has the same number of models as $(A \Leftrightarrow B)$ for any fixed set of proposition symbols that includes A, B, C.

Exercise 4:

Consider a vocabulary with the following symbols:
Occupation(p, o): Predicate. Person p has occupation o.
Customer(p1, p2): Predicate. Person p1 is a customer of person p2.
Boss(p1, p2): Predicate. Person p1 is a boss of person p2.
Doctor, Surgeon, Lawyer, Actor: Constants denoting occupations.
Emily, Joe: Constants denoting people.
Use these symbols to write the following assertions in first-order logic:
a. Emily is either a surgeon or a lawyer.
b. Joe is an actor, but he also holds another job.
c. All surgeons are doctors.
d. Joe does not have a lawyer (i.e., is not a customer of any lawyer).
e. Emily has a boss who is a lawyer.
f. There exists a lawyer all of whose customers are doctors.
g. Every surgeon has a lawyer.

Exercise 5:

For each of the following sentences in English, decide if the accompanying first-order logic sentence is a good translation. If not, explain why not and correct it. (Some sentences may have more than one error!)

a. No two people have the same social security number.

$$\neg\exists x, y, n \;\; Person(x) \wedge Person(y) \;\Rightarrow\; [HasSS\#(x, n) \wedge HasSS\#(y, n)].$$

b. John's social security number is the same as Mary's.

$$\exists n \;\; HasSS\#(John, n) \wedge HasSS\#(Mary, n).$$

c. Everyone's social security number has nine digits.

$$\forall x, n \;\; Person(x) \;\Rightarrow\; [HasSS\#(x, n) \wedge Digits(n, 9)].$$

d. Rewrite each of the above (uncorrected) sentences using a function symbol $SS\#$ instead of the predicate $HasSS\#$.

Exercise 6:

A. Rewrite the following statements in First-Order Logic. Choose appropriate predicates and constants.

1. All dogs howl at night.
2. Anyone who has any cats will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. Tariq has either a cat or a dog.
5. (Conclusion) If Tariq is a light sleeper, then Tariq does not have any mice.

B. Convert the statements in CNF showing all the steps.

C. Apply the Resolution inference rule to prove the conclusion (statement 5) from the first four statements (statements 1 to 4). Make sure that each step all the substitutions are shown as well as the parent clauses and the resolvent.